



FlowState — AI Burnout & Productivity Monitor

A privacy-first, AI-powered burnout detection system using behavioral biometrics — no check-ins, no cloud, no surveillance. Your data never leaves your device.

Burnout Is Silent — And Cumulative



By the time someone realizes they're burnt out, it's already hit. Existing tools fail on three fronts:

Too manual — daily self-reporting that burned-out people skip

Too invasive — employers monitor employees, not help them

Not personalized — compare you to averages, not to yourself

1 in 3 employees globally experience burnout (WHO). Employee burnout costs **\$322B annually** (Gallup).

"Burnout is personal — it's *change from yourself*, not comparison to others." FlowState uses **anomaly detection**, not classification.

Research Backs This

"Prior research shows typing behavior alone can classify 15 distinct emotional states (Epp et al., CHI 2011) and detect cognitive load with 72.39% accuracy (Brizan et al., 2015) — validating behavioral signals as proxies for burnout."

1. Epp et al. classified **15 emotional states** using only keystroke timing — no content captured
2. Brizan et al. reached **72.39% accuracy** on cognitive load detection using typing dynamics alone



SOLUTION

100% Private. 100% Passive.

FlowState passively observes *how* you work — not *what* — builds your personal baseline, and detects deviation. Everything runs **100% on your device**.

Captured (metadata only)

- Time between keypresses
- Backspace / delete rate
- Mouse velocity & click precision
- Scroll speed & patterns
- Window switch frequency

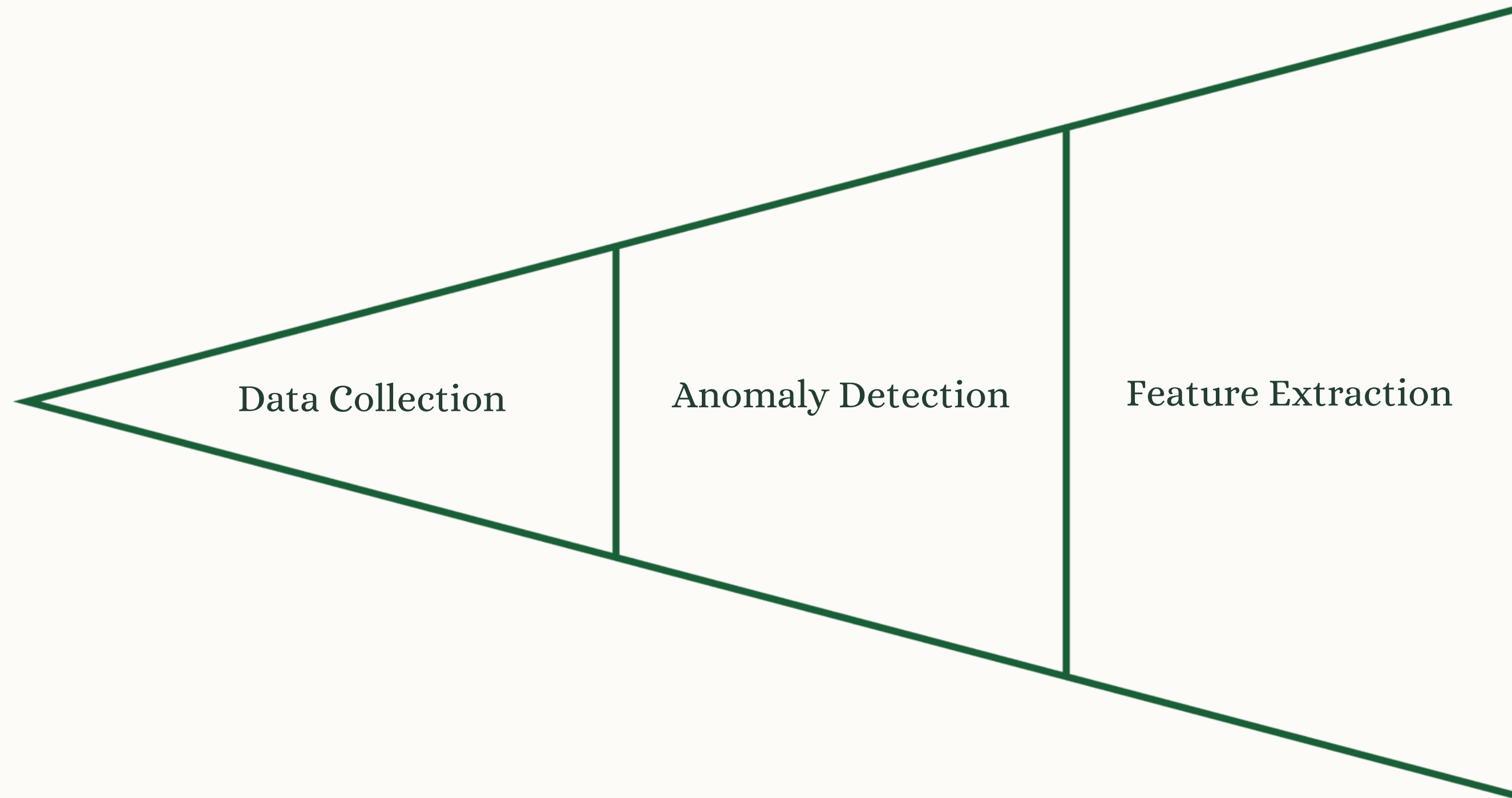
Never captured

- Which keys were pressed
- Any text content or screenshots
- Browsing history or app content
- Anything sent to a server



Raw events discarded after aggregation. Only statistical summaries stored — in a local JSON file only you can access.

Everything Runs On Your Device



No backend. No API. No account. No cloud. Four layers, one device — your behavioral data never leaves your machine.

AI That Learns *You* — Not Everyone Else

Algorithm: Isolation Forest

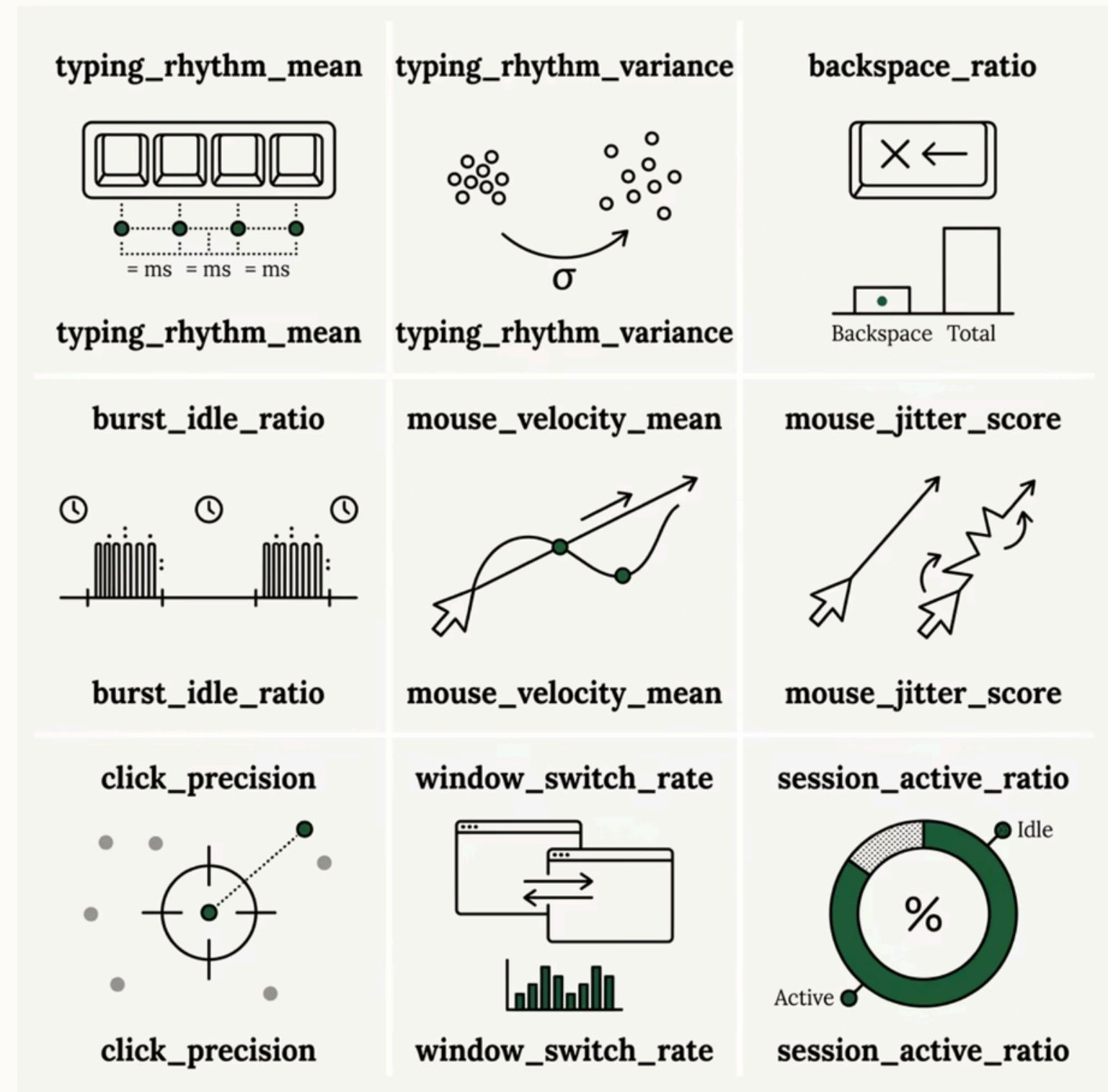
Why: No labeled burnout data needed

How: Learns your personal behavioral baseline

Output: Anomaly score per session (-1 to 0)

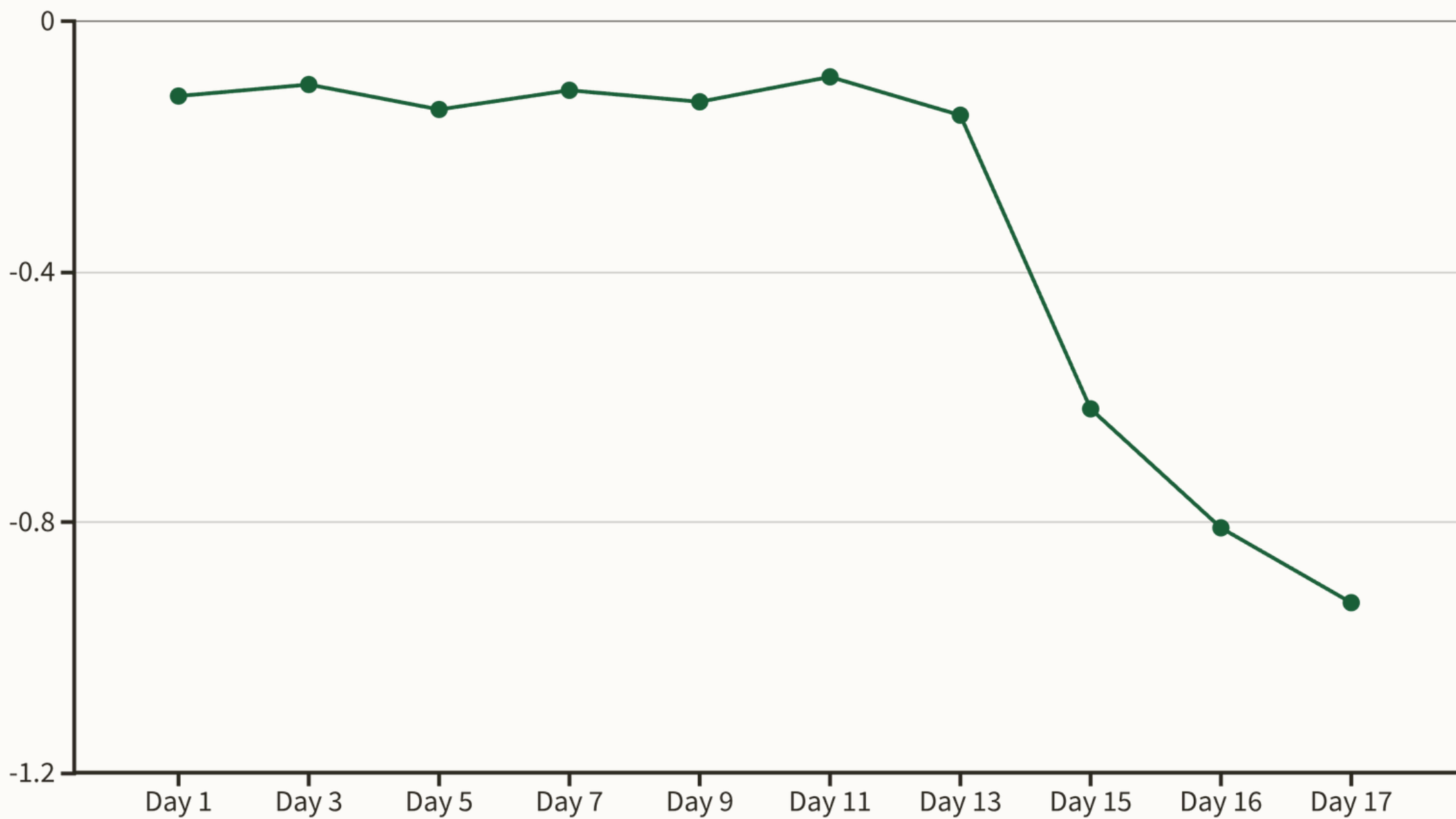
Tech Stack: Python + pynput · pandas · scikit-learn · Local JSON · React + Recharts + Tailwind

📄 **Backend: None.** 100% local.



DEMO

Catching Burnout Before It Breaks You



Days 1–14 establish a **stable green-zone baseline**. Days 15–17 show a sharp spike into the **orange/red critical zone** — behavioral drift detected before the user even noticed.

✔ 100% local — zero cloud dependency

Real-Time Score

Feature Breakdown

Intervention Cards

Real-World Impact & What's Next

Privacy-by-Design

No keystrokes stored. No cloud. No employer access. The opposite of surveillance.

No Labeled Data

Models change from self — not comparison to a dataset.

Completely Passive

No daily forms. Works silently in the background.

Novel Approach

Behavioral AI on metadata — not self-report surveys.



Browser Extension



Wearable Integration



Calendar & GitHub



Federated Learning



Therapist Referral Trigger

Know Before You Break

Existing tools are reactive. FlowGuard is a **passive, privacy-preserving early warning system** — your personal burnout radar that respects your privacy as much as your wellbeing.

100% Local	100% Private	100% Personalized
0% Surveillance	0% Cloud	

 Built for **Technovate 3.0 Hackathon** · MIT License · Ready for live demo